# Exact replication and screening of tropical finiteness certificates for central configurations

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CAOS Research program, `github.com/fsantibanezleal/CAOS_RESEARCH`

2026-08-01
Version 0.04

**Preprint: a machine research record, not peer reviewed.**
Replication-first: the finiteness theorems reproduced here are due to Hampton–Moeckel, Albouy–Kaloshin and Jensen–Leykin, and the equal-mass censuses to Moczurad–Zgliczyński. Our own contributions are the exact instruments, the measured facts and the independent reproductions, each labelled as such. Every number is reproducible from the versioned repository; hypotheses are committed before their runs and refutations are kept in the record.

**Abstract**

We report an exact, fully replayable machine record around Smale's 6th problem (finiteness of planar central configurations). Contributions of this record: (i) an exact calibration of the Albouy–Chenciner mutual-distance equations in open tooling, including a machine rediscovery of their dimension-blindness (the regular tetrahedron coexists with the square in the equal-mass rhombus stratum of the 4-body distance system until the Cayley–Menger equation is adjoined, after which the square, with side $a$ satisfying $32a^6 - 32a^3 + 7 = 0$, is the unique positive point of the stratum); (ii) a machine measurement of the *enrichment law*: adjoining the asymmetric (Roberts) equations and the energy–inertia relation renders the $n = 3$ distance ideal zero-dimensional directly, eliminating the coordinate-line components of the bare symmetric system; (iii) an independent, exact reproduction of the Jensen–Leykin tropical prevariety certificate of generic-mass finiteness for $n = 5$ (arXiv:2301.02305v2): both published $f$-vectors match digit for digit and the pointedness of the recession cone is verified by an independent exact parser; and (iv) a screening study over mass-valuation families and equation-system variants at $n = 4, 5$ which finds two new working valuation families beyond the two published, replicates generic-mass finiteness for $n = 4$ purely polyhedrally in seconds, and shows that removing the ten dependent symmetric Albouy–Chenciner equations destroys the certificate at every tested valuation: the "redundant" equations carry essential tropical information. Every comet of every screened cell is then decided exactly by a rational-arithmetic feasibility certificate, which upgrades the negative half of the screening from absence of a certificate to proof; adding further valid equations is measured separately, showing Dziobek's equations to be tropically active but not decisive at the hard case and the energy–inertia relation to be tropically inert; and two censuses that defeated our own engine are closed by an independent solver with exact cross-verification. All computations are exact (rational or integer arithmetic, or certified polyhedral computation); every stated number is reproducible from committed code and hashes. This is a replication-and-instruments record: the theorems replicated are due to Hampton–Moeckel, Albouy–Kaloshin, and Jensen–Leykin; the measurements in (i), (ii) and (iv) are, to our knowledge, recorded here at this level of explicitness for the first time.

# 1 Introduction

A *central configuration* (CC) of the Newtonian $n$-body problem with masses $m_1, \ldots, m_n > 0$ is a configuration $x = (x_1, \ldots, x_n)$, $x_i \in \mathbb{R}^d$, satisfying

$$\lambda\,(x_i - c) = \sum_{j \neq i} \frac{m_j (x_j - x_i)}{r_{ij}^3}, \qquad i = 1, \ldots, n, \tag{1}$$

for some $\lambda$, where $c$ is the center of mass and $r_{ij} = |x_i - x_j|$. Smale's 6th problem for the 21st century [1] asks, following Chazy [2] and Wintner [3], whether for every $n$ and every choice of positive masses the number of *planar* central configurations, up to symmetry, is finite. The modern ladder is: $n = 4$ finite for all positive masses (Hampton–Moeckel [4]: computer-assisted BKK certificate, count in $[32, 8472]$); $n = 5$ finite except possibly on an explicit codimension-two subvariety of mass space (Albouy–Kaloshin [5]); the spatial 5-body Dziobek case handled generically by Moeckel [6] and explicitly by Hampton–Jensen [7]; $n = 6$ open, reduced by Chang–Chen [8] to a finite list of residual diagram cases; and, for *generic* masses, the recent purely polyhedral certificate of Jensen–Leykin [9], complete through $n = 5$. Rigorous equal-mass censuses for $n \leq 7$ are due to Moczurad–Zgliczyński [10]; with one negative mass a continuum exists (Roberts [11]), so positivity is essential.

This paper is a *machine record*: it documents, with exact artifacts, a replication-first program whose goal is to place every certificate of the ladder inside one open, replayable toolchain, and then to run the experiments the frontier papers themselves call for. We follow a strict discipline: hypotheses are declared and committed before every run; verdicts record refutations as prominently as confirmations; every quantitative claim below is reproducible from the versioned repository, and the git and Zenodo timestamps constitute the priority record of the work.

# 2 Exact calibration of the Albouy–Chenciner system (EXP-001)

Working in the mutual distances $r_{ij}$ with the scale normalization $S_{ij} = r_{ij}^{-3} - 1$ ($S_{ii} = 0$), the symmetric Albouy–Chenciner (AC) equations [12, 4] are, after clearing denominators,

$$f_{ij} = \sum_{k=1}^{n} m_k \left[ S_{ik}\,(r_{jk}^2 - r_{ik}^2 - r_{ij}^2) + S_{jk}\,(r_{ik}^2 - r_{jk}^2 - r_{ij}^2) \right] = 0. \tag{2}$$

Our exact implementation (sympy over $\mathbb{Q}$) reproduces, for four exact mass vectors, the classical $n = 3$ landscape: the Lagrange point $r_{12} = r_{13} = r_{23} = 1$ identically in symbolic masses, and exactly one positive collinear solution per ordering (Euler–Moulton; for equal masses the 2-middle chart value is $r_{12} = r_{23} = \sqrt[3]{90}/6$), with the degree-54 symbolic Euler eliminant persisted. For $n = 4$ each cleared AC polynomial has exactly 14 monomials, matching the count stated in [4].

The instructive result of the calibration is a machine *refutation* of a declared prediction. In the equal-mass rhombus stratum $r_{12} = r_{23} = r_{34} = r_{14} = a$, $r_{13} = r_{24} = b$ of $n = 4$, the bare AC system has *two* positive solutions: the planar square, with $b^2 = 2a^2$ and $a^3 = (4 + \sqrt{2})/8$ (minimal polynomial $32a^6 - 32a^3 + 7$), and the regular tetrahedron $a = b = 1$, whose bordered Cayley–Menger determinant equals $4 \neq 0$. The distance equations are dimension-blind, exactly as stated in [4]; planarity is restored only by adjoining the Cayley–Menger equation, after which the square is the unique positive point of the stratum. We record the scale-free invariant $J = UI^{1/2}/M^{5/2}$ at both anchors: $J_{\square} = \frac{1}{16} + \frac{\sqrt{2}}{8}$, $J_{\text{tet}} = \frac{3\sqrt{6}}{32}$.

Two toolchain findings of independent interest were caught and are now regression-gated in the repository: `sympy`'s `solve_poly_system` silently returns incomplete solution lists on these systems (it missed the square, whose eliminant factor $4b^6 - 4b^3 - 7$ is even radical-solvable), and `is_positive` returns `None` on nested algebraic numbers, so naive filters silently drop genuine solutions. Verdict-grade counting in our pipeline therefore uses univariate eliminants that are provably elements of the ideal (lex Gröbner elimination, or Stickelberger multiplication-matrix characteristic polynomials), complete real root isolation, and exact residual acceptance.

## 3 The enrichment law (EXP-002)

The bare symmetric system is never zero-dimensional: the line $r_{13} = r_{23} = 0$ lies in its variety for all masses (machine-verified symbolically), and saturation is expensive. Adjoining the $n(n-1)$ asymmetric equations $g_{ij} = \sum_k m_k S_{ik}(r_{jk}^2 - r_{ik}^2 - r_{ij}^2)$ (observed by G. Roberts; see [7, 9]) and the energy–inertia relation $U = MI$ changes the picture completely: for all four sample mass vectors the enriched ideal is zero-dimensional *directly* (Gröbner basis of size 24 in under a second each), and the full positive censuses, where decided, are exactly classical: four points (one equilateral, three collinear), with no spurious solutions. The planar rhombus census on the enriched system with Cayley–Menger adjoined yields the square alone. This is the affine-level instance of a pattern that recurs at every level of the subject: Hampton–Moeckel could not run their method on the six AC equations alone and required Dziobek's equations [4]; Hampton–Jensen kept the asymmetric equations to refine their tropical cones [7]; and Jensen–Leykin explicitly call for "different versions of the Albouy–Chenciner equations" at $n = 6$ [9]. Section 5 quantifies this pattern at the prevariety level.

## 4 Exact reproduction of the Jensen–Leykin $n = 5$ certificate (EXP-003)

Jensen–Leykin [9] prove generic-mass finiteness of planar CCs for $n \leq 5$ by a purely polyhedral computation: specialize the masses into the Puiseux field with chosen valuations, compute the tropical prevariety of the system (20 asymmetric AC + 10 symmetric AC + 5 planar Cayley–Menger polynomials for $n = 5$) with `gfan` 0.7, and verify that every connected component has a pointed recession cone; Bieri–Groves and the density of tropicalization fibers then yield finiteness for almost all masses.

We rebuilt `gfan` 0.7 from the author tarball (one two-line `#include <cstdint>` patch for gcc 13; tarball and binary SHA-256 recorded) and ran the paper's own command pipeline. Results: with valuations $(1, 3, 9, 27, 81)$ the prevariety $f$-vector is $(1506, 4744, 8586, 8787, 4652, 993)$, and among its 85 unbounded ray directions none has a positive coordinate (verified by our independent exact parser): the globally pointed certificate, exactly as published. With valuations $(1, 4, 9, 16, 25)$ the $f$-vector is $(3586, 12012, 18531, 15625, 7072, 1357)$, again matching [9] digit for digit. Each run took under six minutes of wall time on 30 threads ($\approx 145$ cpu-minutes). To our knowledge this is the first published independent replication of the [9] computation; at the measured throughput, a single $n = 6$ attempt at the scale they report ($\approx 100$ cpu-days) costs about three wall-days on commodity hardware.

## 5 Valuation and equation-variant screening (EXP-004)

The certificate of [9] has two designed inputs: the mass valuations and the equation system. The paper publishes two working valuation choices for $n = 5$ and, for the inconclusive $n = 6$ attempt, asks explicitly for "more experimentation with different valuations or different versions of the Albouy–Chenciner equations". We screened both axes on a 16-cell grid: two systems (S1, the full set of asymmetric + symmetric AC equations + planar Cayley–Menger determinants; S2, the same with the $\binom{n}{2}$ dependent symmetric equations removed) crossed with six valuation families at $n = 5$, plus four cells at $n = 4$. Pointedness is decided per connected component (*comet*) by an exact certificate: a rational vector $c$ with $c \cdot r < 0$ for every recession generator $r$, verified in rational arithmetic (a floating-point solver only proposes $c$).

| Valuations at $n = 5$ (system S1) | comets | pointed | status |
|---|---|---|---|
| $(1, 3, 9, 27, 81)$ powers of 3 | 281 | 281 | certificate (globally pointed) |
| $(1, 4, 9, 16, 25)$ squares | 257 | 257 | certificate |
| $(1, 2, 4, 8, 16)$ powers of 2 | 266 | 266 | certificate (new) |
| $(2, 3, 5, 7, 11)$ primes | 250 | 250 | certificate (new) |
| $(0, 1, 2, 3, 4)$ arithmetic | 181 | 180 | no certificate (1 comet undecided) |
| $(1, 1, 9, 27, 81)$ repeated | 211 | 210 | no certificate (1 comet undecided) |

Table 1: Comet-level screening at $n = 5$. The squares row reproduces the 257 components reported in [9], a quantity independent of the $f$-vector, which validates the component extraction as well as the polyhedral computation.

Three findings.

*(a) Two new working valuation families.* Powers of two and primes both certify at $n = 5$; before this screening, two families were known to work. Since the certificate is a statement about one point of the valuation space, each working family is an independent verification of generic finiteness for $n = 5$, and the enlarged pool is what makes an $n = 6$ attempt a design choice rather than a lottery. This matters operationally: powers of three reach $t^{243}$ at $n = 6$ and exceed `gfan`'s machine-integer fast path (our first $n = 6$ attempt aborted on a detected overflow after six minutes), whereas powers of two reach only $t^{32}$.

*(b) The dependent symmetric equations are essential.* Every S2 cell has a strictly larger $f$-vector in every entry than its S1 counterpart and yields no certificate at any valuation, including the one where S1 is globally pointed; the single undecided comet grows from 71–95 recession generators in the S1 controls to 241–2224 in S2. The equations that are algebraically redundant (each symmetric equation is the sum of two asymmetric ones) are tropically load-bearing. This is the prevariety-level form of the affine phenomenon of Section 3, and it agrees with the historical record: [4] could not run their method on the six AC equations alone, and [7] kept the redundant family deliberately.

*(c) $n = 4$ falls in seconds, and the hard case is exactly the predicted one.* With valuations $(1, 3, 9, 27)$ and $(1, 2, 4, 8)$ the $n = 4$ prevariety is globally pointed (10 comets each; 4 and 2 seconds), a purely polyhedral replication of generic-mass finiteness for the four-body problem; arithmetic valuations $(0, 1, 2, 3)$ also certify (9 comets). The equal-valuation case $(0, 0, 0, 0)$ leaves a single undecided comet with 49 generators, matching the observation of [9] that this specialization is equivalent in difficulty to the original computation of [4].

### 5.1 Every comet decided exactly (EXP-007)

The screening above was reported with an explicit weakness: on the failing controls the heuristic returned neither certificate, so those cells read “no certificate” rather than “not pointed”. We closed that gap. A cone is pointed exactly when the only nonnegative solution of $\sum_i \lambda_i g_i = 0$ is $\lambda = 0$, which is a single linear feasibility question; we decide it with a phase-I simplex in exact rational arithmetic (Bland’s rule, so termination is guaranteed). Feasibility returns an explicit nonnegative zero combination, infeasibility returns a separating vector, and both are re-verified by exact substitution outside the decider.

All sixteen cells are now decided in 33 seconds total, with zero certificate-verification failures. Every comet previously certified pointed is confirmed (no flips), and each failing control has exactly one comet that is provably UNPOINTED: at $n = 5$ the arithmetic valuations $(0, 1, 2, 3, 4)$ (180 pointed, one unpointed) and the repeated valuations, and at $n = 4$ the equal valuations. Our data therefore independently confirms, rather than merely being consistent with, the failure of arithmetic valuations reported in [9], and localises it to a single component out of 181.

We keep the logical direction straight: pointedness of every comet is a *sufficient* condition for dimension zero, so an unpointed comet proves only that this certificate route does not close, never that finiteness fails. The prevariety over-approximates the tropical variety.

### 5.2 Adding equations: what helps and what does not (EXP-008)

Since removing the redundant symmetric equations is catastrophic, the converse is worth measuring. At $n = 4$, where the planar stratum *is* the Dziobek stratum and Dziobek’s equations are therefore available (they are what Hampton–Moeckel [4] needed when the Albouy–Chenciner equations alone defeated their method), we ran the baseline against three enrichments.

Two declared predictions failed, informatively. First, we predicted that adding equations shrinks the $f$-vector; it does at equal valuations (a 25 percent reduction) but GROWS it at powers of two and at arithmetic valuations. The prediction confused the prevariety as a set with the subdivision that represents it: intersecting more tropical hypersurfaces can only shrink the set while refining the fan, so cone counts may rise. The invariant that behaves monotonically in our data is the comet count, which falls from 10 to 7 and from 9 to 6 when Dziobek is adjoined. Second, we predicted that Dziobek’s equations would rescue the hard equal-valuation case. They do not: the prevariety shrinks by a quarter and its single comet stays provably unpointed. The enrichment that rescued the *algebraic* proof therefore does not rescue the *tropical* certificate at the same specialisation, which separates two roles that are easy to conflate.

By contrast the energy–inertia relation is tropically inert: it reproduces the baseline $f$-vector exactly at two of the three valuations. Dependence in the ideal predicts tropical relevance in neither direction, which is the sharper form of the lesson above, and it lets us drop that equation from expensive runs at no cost.

## 6 Censuses across two independent engines (EXP-006)

Two of the $n = 3$ censuses had been left inconclusive at our own engine’s budget. We recomputed them with msolve (Gröbner bases, rational univariate representation and certified real-root isolation), keeping our exact machinery as the verification layer: msolve returns rational isolating boxes, and containment of an exact algebraic point in a rational box is decided exactly. Both open cells close in under a second each, and for all four mass vectors the positive census is exactly the classical answer, one equilateral point and one collinear point per ordering, with every one of our independently computed exact points inside a distinct box and no unexplained boxes. Two implementations that share no code agree on every decided instance.

The same experiment refuted its own third prediction. We expected the enriched planar $n = 4$ system to admit an isolated census matching the published equal-mass class count; msolve reports the affine system to have dimension one, so there is no isolated census to compare. Diagnostics exclude the vanishing-distance loci as the cause, which leaves the reading the literature already acts on: the census belongs in the torus, where all mutual distances are invertible, which is precisely the setting of [4]. We record the count as untested rather than quoting the published value as if we had reproduced it.

Following that lesson, we attempted the $n = 4$ equal-mass census where it belongs, in the torus, by two independent exact formulations: the enriched planar system under a Rabinowitsch saturation of the distance product (seven unknowns) and the z-system of [4] (twelve unknowns, the mutual distances joined by the Dziobek variables). The exact square of Section 2 passes the smoke test of the first system by exact residual substitution. Neither formulation produced any msolve output within a declared and mechanically enforced budget of 3600 seconds per route (EXP-009), so the published equal-mass counts remain untested by us. We note that the literature contains, to our knowledge, no direct algebraic solve of this census either: [4] bound the count by a mixed volume rather than solving, and the rigorous equal-mass counts of [10] come from interval methods. The measured cliff between $n = 3$ (both censuses close in under a second) and $n = 4$ (no output in an hour, twice) redirects our counting work to the bounding rung and our finiteness work to dimension computations, as follows.

**The incidence-dimension lane (EXP-010, declared and launched).** The object is the Dziobek variety of $n = 4$: the vanishing of the cleared Dziobek differences $S_{12}S_{34} = S_{13}S_{24} = S_{14}S_{23}$ together with the planar Cayley–Menger determinant, in the torus of the six mutual distances. Masses do not appear in this formulation, and generic finiteness corresponds to this variety having the dimension of the projectivized mass space, three. The instrument keeps every algebraic call zero-dimensional by construction: recorded random integer linear sections of codimension three (which should meet the variety in finitely many points, whose number is degree data) and codimension four (which should miss it), decided by msolve under caps, with a deterministic Gröbner staircase rung run separately. The declared smoke gate discriminates three exact anchors in one second: the exact square of Section 2 lies on the variety (certified by polynomial reduction modulo $8a^3 = 4 + \sqrt{2}$ written as two polynomial relations, no radical heuristics), the unit tetrahedron satisfies the Dziobek equations but is excluded by Cayley–Menger, and the 3-4-5 rectangle satisfies Cayley–Menger but violates the Dziobek equations. Results will be reported with the drawn sections verbatim in the next version of this record.

# 7 The $n = 6$ attempt (EXP-005, in progress)

The screening of Section 5 turns the $n = 6$ question into a design problem with two dials. What we have measured since is a *tooling* barrier, characterized on four attempts with `gfan` 0.7. On the machine-integer fast path the computation aborts within minutes on a *detected* overflow (the benign failure mode: `gfan` raises rather than returning a wrong answer), and it does so for powers of three and for powers of two alike, although the latter caps the valuation exponents at 32 instead of 243, so the failure is not a matter of valuation magnitude. On the arbitrary-precision path, the powers-of-two run died after fifteen wall-hours (about six cpu-days) on an internal assertion in the hypersurface-intersection code. Version 0.7 therefore fails at $n = 6$ in both arithmetic modes, in two distinct ways, while the same binary reproduces the published $n = 5$ certificates exactly; [9] report their own $n = 6$ computation as completing (and inconclusive on pointedness), so their run differed in version, flags, or valuation. We rebuilt the pipeline on the 0.8beta release, validated it against our $n = 5$ control (exact f-vector agreement), and confirmed that it clears 0.7's first failure point. Two 0.8beta runs are in progress with checkpointing (powers of two on the fast path; powers of three with arbitrary-precision integers). The outcome, positive or negative, will be reported with its exact certificates in the next version of this record; a positive outcome would extend generic-mass finiteness to $n = 6$, and a negative one localizes the obstruction to specific comets, data that has not been published for any failed attempt.

## 8 Honesty ledger

This record is replication-first. The finiteness theorems replicated are due to Hampton–Moeckel, Albouy–Kaloshin, and Jensen–Leykin; the equal-mass censuses to Moczurad–Zgliczyński. Our measurements (the dimension-blindness anchors with their exact minimal polynomials and $J$-invariants; the enrichment law at affine and prevariety level; the screening tables) are machine facts about known objects, stated with their complete provenance. Open items are tracked in the repository backlog; in particular, censuses for two integer-separated mass vectors are inconclusive under our current sympy engine caps and are being re-run under an independent engine (`msolve`) with the sympy layer retained as verification. No claim in this paper depends on an unverified literature statement; the repository flags every such statement as UNVERIFIED until read from its primary source.

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